

Final Project

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Read in the cogimp0 dataset

```
# Access and read the cogimp0 file from the drive
cogimp0 = drive_download(
  as_id("1xDBY_8-djIWfKWTBKrx1YbZ9mkw6z1J2"),
  path = tempfile(fileext = ".sav"),
  overwrite = TRUE
) %>%
  { read_sav(.$local_path) }

## File downloaded:
## * 'COGIMP9220A_R.sav' <id: 1xDBY_8-djIWfKWTBKrx1YbZ9mkw6z1J2>
## Saved locally as:
## * '/var/folders/_d/fx514zw972j96qby8l86rsgh0000gn/T//Rtmp32VfN6/file17bc97d2b6c40.sav'
# generate codebook dataframe
cogimp0_codebook = labelled::generate_dictionary(cogimp0)
```

Read in the 2022 tracker file dataset

```
# read in the trk2022v2 file
trk2022v2 = drive_download(
  as_id("1G0wZriMuTMftZYP834aJG7nMrN3k4T5E"),
  path = tempfile(fileext = ".sav"),
  overwrite = TRUE
) %>%
  { read_sav(.$local_path) }

## File downloaded:
## * 'trk2022tr_r.sav' <id: 1G0wZriMuTMftZYP834aJG7nMrN3k4T5E>
## Saved locally as:
## * '/var/folders/_d/fx514zw972j96qby8l86rsgh0000gn/T//Rtmp32VfN6/file17bc96ddf08c.sav'
# generate codebook
trk2022v2_codebook = labelled::generate_dictionary(trk2022v2)
```

Combine the data and variables we need

```
cogimp0 <- cogimp0 %>% mutate(hhid = as.character(HHID), pn = as.character(PN))
trk2022v2 <- trk2022v2 %>% mutate(hhid = as.character(hhid), pn = as.character(pn))

data_w13 <- cogimp0 %>%
  dplyr::select(hhid, pn, R13M0, R13DY, R13YR, R13DW, R13SCIS, R13CACT, R13PRES, R13VP) %>% rename(
    Month= R13M0,
    Day= R13DY,
    Year= R13YR,
    DOW= R13DW,
    Scissors= R13SCIS,
    Cactus= R13CACT,
    President= R13PRES,
    VP = R13VP
  ) %>%
  mutate(wave = "W13")

data_w3 <- cogimp0 %>%
  dplyr::select(hhid, pn, R3M0, R3DY, R3YR, R3DW, R3SCIS, R3CACT, R3PRES, R3VP) %>%
  rename(
    Month= R3M0,
    Day= R3DY,
    Year= R3YR,
    DOW = R3DW,
    Scissors = R3SCIS,
    Cactus = R3CACT,
    President = R3PRES,
    VP = R3VP
  ) %>%
  mutate(wave = "W3")

# data_long <- bind_rows(data_w3, data_w13)

cov_w13 <- trk2022v2 %>%
  dplyr::select(hhid, pn, page, gender, degree, race) %>%
  rename(age = page) %>%
  mutate(wave = "W13")

cov_w3 <- trk2022v2 %>%
  dplyr::select(hhid, pn, eage, gender, degree, race) %>%
  rename(age = eage) %>%
  mutate(wave = "W3")

# covs_long <- bind_rows(cov_w3, cov_w13)

# lca_raw <- data_long %>%
#   inner_join(covs_long, by = c("hhid", "pn", "wave"))

# indicator_names <- c("Month", "Day", "Year", "DOW", "Scissors", "Cactus", "President", "VP")

lca_w3 <- data_w3 %>%
  inner_join(cov_w3, by = c("hhid", "pn", "wave"))

indicator_names <- c("Month", "Day", "Year", "DOW", "Scissors", "Cactus", "President", "VP")
```

```
#lca_ready <- lca_raw %>%
#   dplyr::mutate(across(all_of(indicator_names),
#   ~ ifelse(is.na(.), NA, as.integer(. + 1))))
```

```
lca_ready_w3 <- lca_w3 %>%
  dplyr::mutate(across(all_of(indicator_names),
    ~ ifelse(is.na(.), NA, as.integer(. + 1))))
```

```
summary(lca_w3[, indicator_names])
```

```
##      Month      Day      Year      DOW
##  Min.   :0.000  Min.   :0.000  Min.   :0.000  Min.   :0.000
## 1st Qu.:1.000  1st Qu.:1.000  1st Qu.:1.000  1st Qu.:1.000
## Median :1.000  Median :1.000  Median :1.000  Median :1.000
## Mean   :0.974  Mean   :0.835  Mean   :0.975  Mean   :0.976
## 3rd Qu.:1.000  3rd Qu.:1.000  3rd Qu.:1.000  3rd Qu.:1.000
## Max.   :1.000  Max.   :1.000  Max.   :1.000  Max.   :1.000
## NA's   :24690  NA's   :24690  NA's   :24690  NA's   :24690
##      Scissors      Cactus      President      VP
##  Min.   :0.000  Min.   :0.000  Min.   :0.000  Min.   :0.000
## 1st Qu.:1.000  1st Qu.:1.000  1st Qu.:1.000  1st Qu.:0.000
## Median :1.000  Median :1.000  Median :1.000  Median :1.000
## Mean   :0.988  Mean   :0.894  Mean   :0.938  Mean   :0.711
## 3rd Qu.:1.000  3rd Qu.:1.000  3rd Qu.:1.000  3rd Qu.:1.000
## Max.   :1.000  Max.   :1.000  Max.   :1.000  Max.   :1.000
## NA's   :24690  NA's   :24690  NA's   :24690  NA's   :24690
```

```
summary(lca_w3$age)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      21.0   67.0   999.0   751.3   999.0   999.0
```

```
table(lca_w3$gender, useNA = "ifany")
```

```
##
##      1      2
## 17830 23211
```

```
table(lca_w3$degree, useNA = "ifany")
```

```
##
##      0      1      2      3      4      5      6      9
## 10117 2015 17600 2041 4771 2418 797 1282
```

```
table(lca_w3$race, useNA = "ifany")
```

```
##
##      0      1      2      7
## 111 29627 7871 3432
```

```
table(lca_w3$wave)
```

```
##
##      W3
## 41041
```

Fit LCA model

Without covariates

```
set.seed(687)
f_ind <- cbind(Month, Day, Year, DOW, Scissors, Cactus, President, VP) ~ 1
sub_df <- lca_ready_w3 %>% dplyr::sample_frac(0.5)
fit_list <- lapply(2:5, function(k) {
  polCA(f_ind, data = sub_df, nclass = k, maxiter = 1500, nrep = 3, verbose = FALSE)
})

data.frame(
  k = 2:5,
  AIC = sapply(fit_list, \(m) m$aic),
  BIC = sapply(fit_list, \(m) m$bic),
  Gsq = sapply(fit_list, \(m) m$Gsq)
)
```

```
##   k      AIC      BIC      Gsq
## 1 2 30771.76 30890.96 683.0069
## 2 3 30435.14 30617.44 328.3878
## 3 4 30328.27 30573.67 203.5165
## 4 5 30309.01 30617.52 166.2595
```

We will use 4 classes with BIC the lowest for w3.

```
set.seed(687)
lca_k4_w3 <- polCA(f_ind, data = lca_ready_w3, nclass = 4, maxiter = 5000, nrep = 20, verbose = TRUE)

## Model 1: llik = -30487.05 ... best llik = -30487.05
## Model 2: llik = -30491.71 ... best llik = -30487.05
## Model 3: llik = -30487.05 ... best llik = -30487.05
## Model 4: llik = -30487.05 ... best llik = -30487.05
## Model 5: llik = -30487.05 ... best llik = -30487.05
## Model 6: llik = -30487.05 ... best llik = -30487.05
## Model 7: llik = -30491.71 ... best llik = -30487.05
## Model 8: llik = -30487.05 ... best llik = -30487.05
## Model 9: llik = -30487.05 ... best llik = -30487.05
## Model 10: llik = -30487.05 ... best llik = -30487.05
## Model 11: llik = -30487.05 ... best llik = -30487.05
## Model 12: llik = -30487.05 ... best llik = -30487.05
## Model 13: llik = -30487.05 ... best llik = -30487.05
## Model 14: llik = -30487.05 ... best llik = -30487.05
## Model 15: llik = -30487.05 ... best llik = -30487.05
## Model 16: llik = -30487.05 ... best llik = -30487.05
## Model 17: llik = -30487.05 ... best llik = -30487.05
## Model 18: llik = -30487.05 ... best llik = -30487.05
## Model 19: llik = -30491.71 ... best llik = -30487.05
## Model 20: llik = -30487.05 ... best llik = -30487.05
## Conditional item response (column) probabilities,
## by outcome variable, for each class (row)
##
## $Month
##           Pr(1) Pr(2)
## class 1:  0.6334 0.3666
## class 2:  0.0055 0.9945
```

```

## class 3: 0.0292 0.9708
## class 4: 0.1958 0.8042
##
## $Day
##           Pr(1) Pr(2)
## class 1: 0.9003 0.0997
## class 2: 0.1058 0.8942
## class 3: 0.2800 0.7200
## class 4: 0.6655 0.3345
##
## $Year
##           Pr(1) Pr(2)
## class 1: 0.8287 0.1713
## class 2: 0.0019 0.9981
## class 3: 0.0369 0.9631
## class 4: 0.1428 0.8572
##
## $DOW
##           Pr(1) Pr(2)
## class 1: 0.4647 0.5353
## class 2: 0.0048 0.9952
## class 3: 0.0312 0.9688
## class 4: 0.2186 0.7814
##
## $Scissors
##           Pr(1) Pr(2)
## class 1: 0.0865 0.9135
## class 2: 0.0065 0.9935
## class 3: 0.0264 0.9736
## class 4: 0.0354 0.9646
##
## $Cactus
##           Pr(1) Pr(2)
## class 1: 0.6211 0.3789
## class 2: 0.0491 0.9509
## class 3: 0.3399 0.6601
## class 4: 0.1829 0.8171
##
## $President
##           Pr(1) Pr(2)
## class 1: 0.7079 0.2921
## class 2: 0.0109 0.9891
## class 3: 0.2893 0.7107
## class 4: 0.0000 1.0000
##
## $VP
##           Pr(1) Pr(2)
## class 1: 0.9539 0.0461
## class 2: 0.1626 0.8374
## class 3: 0.8974 0.1026
## class 4: 0.2877 0.7123
##
## Estimated class population shares
## 0.0147 0.7973 0.1494 0.0385

```

```
##
## Predicted class memberships (by modal posterior prob.)
## 0.0143 0.8428 0.126 0.0169
##
## =====
## Fit for 4 latent classes:
## =====
## number of observations: 16351
## number of estimated parameters: 35
## residual degrees of freedom: 220
## maximum log-likelihood: -30487.05
##
## AIC(4): 61044.1
## BIC(4): 61313.67
## G^2(4): 289.0484 (Likelihood ratio/deviance statistic)
## X^2(4): 446.6359 (Chi-square goodness of fit)
##
```

```
lca_k4_w3$probs
```

```
## $Month
##           Pr(1)      Pr(2)
## class 1: 0.633396914 0.3666031
## class 2: 0.005508788 0.9944912
## class 3: 0.029248199 0.9707518
## class 4: 0.195826123 0.8041739
##
## $Day
##           Pr(1)      Pr(2)
## class 1: 0.9003198 0.09968016
## class 2: 0.1057820 0.89421798
## class 3: 0.2799834 0.72001660
## class 4: 0.6654896 0.33451038
##
## $Year
##           Pr(1)      Pr(2)
## class 1: 0.828735546 0.1712645
## class 2: 0.001891506 0.9981085
## class 3: 0.036864284 0.9631357
## class 4: 0.142788567 0.8572114
##
## $DOW
##           Pr(1)      Pr(2)
## class 1: 0.464695278 0.5353047
## class 2: 0.004776264 0.9952237
## class 3: 0.031208886 0.9687911
## class 4: 0.218567216 0.7814328
##
## $Scissors
##           Pr(1)      Pr(2)
## class 1: 0.086456894 0.9135431
## class 2: 0.006483089 0.9935169
## class 3: 0.026353363 0.9736466
## class 4: 0.035395135 0.9646049
##
```

```
## $Cactus
##           Pr(1)      Pr(2)
## class 1: 0.62110589 0.3788941
## class 2: 0.04907937 0.9509206
## class 3: 0.33994877 0.6600512
## class 4: 0.18292733 0.8170727
##
## $President
##           Pr(1)      Pr(2)
## class 1: 7.079249e-01 0.2920751
## class 2: 1.089080e-02 0.9891092
## class 3: 2.893035e-01 0.7106965
## class 4: 3.740145e-12 1.0000000
##
## $VP
##           Pr(1)      Pr(2)
## class 1: 0.9538861 0.04611392
## class 2: 0.1626035 0.83739647
## class 3: 0.8974073 0.10259266
## class 4: 0.2877396 0.71226043

lca_ready_w3$rowid <- seq_len(nrow(lca_ready_w3))
mf <- model.frame(f_ind, data = lca_ready_w3, na.action = na.omit)
idx <- as.integer(row.names(mf))
tab_w3 <- table(lca_k4_w3$predclass, lca_ready_w3$wave[idx])
tab_w3
```

```
##
##      W3
##  1   234
##  2 13780
##  3  2060
##  4   277
```

With covariates

```
set.seed(687)
f_cov <- cbind(Month, Day, Year, DOW, Scissors, Cactus, President, VP) ~
  age + factor(gender) + factor(degree) + factor(race)

lca_cov_w3 <- poLCA(f_cov, data = lca_ready_w3, nclass = 4, maxiter = 5000, nrep = 20, verbose = TRUE)

## Model 1: llik = -29756.03 ... best llik = -29756.03
## Model 2: llik = -29107.13 ... best llik = -29107.13
## Model 3: llik = -31303.82 ... best llik = -29107.13
## Model 4: llik = -29601.15 ... best llik = -29107.13
## Model 5: llik = -29500.71 ... best llik = -29107.13
## Model 6: llik = -32725.86 ... best llik = -29107.13
## Model 7: llik = -28977.06 ... best llik = -28977.06
## Model 8: llik = -29815.7 ... best llik = -28977.06
## Model 9: llik = -29402.7 ... best llik = -28977.06
## Model 10: llik = -29732.03 ... best llik = -28977.06
## Model 11: llik = -33222.14 ... best llik = -28977.06
## Model 12: llik = -30783.98 ... best llik = -28977.06
## Model 13: llik = -29768.93 ... best llik = -28977.06
## Model 14: llik = -29815.36 ... best llik = -28977.06
```

```

## Model 15: llik = -29092.85 ... best llik = -28977.06
## Model 16: llik = -29315.02 ... best llik = -28977.06
## Model 17: llik = -29816.78 ... best llik = -28977.06
## Model 18: llik = -30136.63 ... best llik = -28977.06
## Model 19: llik = -29449.8 ... best llik = -28977.06
## Model 20: llik = -30328.53 ... best llik = -28977.06
## Conditional item response (column) probabilities,
## by outcome variable, for each class (row)
##
## $Month
##           Pr(1) Pr(2)
## class 1: 0.0186 0.9814
## class 2: 0.0217 0.9783
## class 3: 0.0091 0.9909
## class 4: 0.4466 0.5534
##
## $Day
##           Pr(1) Pr(2)
## class 1: 0.1971 0.8029
## class 2: 0.2184 0.7816
## class 3: 0.1129 0.8871
## class 4: 0.8006 0.1994
##
## $Year
##           Pr(1) Pr(2)
## class 1: 0.0095 0.9905
## class 2: 0.0189 0.9811
## class 3: 0.0034 0.9966
## class 4: 0.6113 0.3887
##
## $DOW
##           Pr(1) Pr(2)
## class 1: 0.0177 0.9823
## class 2: 0.0188 0.9812
## class 3: 0.0095 0.9905
## class 4: 0.3954 0.6046
##
## $Scissors
##           Pr(1) Pr(2)
## class 1: 0.0072 0.9928
## class 2: 0.0244 0.9756
## class 3: 0.0067 0.9933
## class 4: 0.0820 0.9180
##
## $Cactus
##           Pr(1) Pr(2)
## class 1: 0.0000 1.0000
## class 2: 0.4670 0.5330
## class 3: 0.0274 0.9726
## class 4: 0.5315 0.4685
##
## $President
##           Pr(1) Pr(2)
## class 1: 0.1329 0.8671

```



```

## class 2: 0.1432 0.8568
## class 3: 0.0000 1.0000
## class 4: 0.5239 0.4761
##
## $VP
##           Pr(1) Pr(2)
## class 1: 0.6767 0.3233
## class 2: 0.6837 0.3163
## class 3: 0.0512 0.9488
## class 4: 0.8343 0.1657
##
## Estimated class population shares
## 0.1846 0.1571 0.6292 0.0292
##
## Predicted class memberships (by modal posterior prob.)
## 0.1814 0.131 0.6623 0.0253
##
## =====
## Fit for 4 latent classes:
## =====
## 2 / 1
##           Coefficient Std. error t value Pr(>|t|)
## (Intercept)      1.67748    0.08022  20.911   0.000
## age              0.00046    0.00012   3.728   0.000
## factor(gender)2  -0.20341    0.12064  -1.686   0.093
## factor(degree)1  -1.91330    0.25302  -7.562   0.000
## factor(degree)2  -1.35427    0.12512 -10.824   0.000
## factor(degree)3  -3.96471    0.00641 -618.732   0.000
## factor(degree)4  -3.57331    0.03408 -104.860   0.000
## factor(degree)5  -1.98295    0.05564  -35.638   0.000
## factor(degree)6  -3.17634    0.02425 -130.975   0.000
## factor(degree)9  -0.00823    0.00047  -17.404   0.000
## factor(race)1    -2.08660    0.08626  -24.190   0.000
## factor(race)2     2.94052    0.05485  53.607   0.000
## factor(race)7     0.83652    0.10000   8.365   0.000
## =====
## 3 / 1
##           Coefficient Std. error t value Pr(>|t|)
## (Intercept)      1.31221    0.06949  18.884   0.000
## age             -0.00008    0.00007  -1.091   0.277
## factor(gender)2  -0.63648    0.07672  -8.296   0.000
## factor(degree)1   0.21833    0.12897   1.693   0.092
## factor(degree)2   0.91680    0.07747  11.834   0.000
## factor(degree)3   1.13657    0.18677   6.085   0.000
## factor(degree)4   1.26058    0.11613  10.855   0.000
## factor(degree)5   1.42627    0.14048  10.153   0.000
## factor(degree)6   1.71164    0.11157  15.342   0.000
## factor(degree)9  -0.02524    0.00058 -43.521   0.000
## factor(race)1    -0.46454    0.06862  -6.770   0.000
## factor(race)2     1.73319    0.06030  28.741   0.000
## factor(race)7     0.06234    0.10153   0.614   0.540
## =====
## 4 / 1
##           Coefficient Std. error t value Pr(>|t|)

```

```
## (Intercept)      -0.01810      0.10086     -0.179      0.858
## age              0.00165      0.00017      9.936      0.000
## factor(gender)2  -0.68966      0.14261     -4.836      0.000
## factor(degree)1  -1.89818      0.10439    -18.183      0.000
## factor(degree)2  -1.44939      0.15239     -9.511      0.000
## factor(degree)3  -2.20810      0.02155   -102.471      0.000
## factor(degree)4  -2.59499      0.05265    -49.287      0.000
## factor(degree)5  -1.35554      0.03156    -42.948      0.000
## factor(degree)6  -1.91595      0.01224   -156.508      0.000
## factor(degree)9   0.01827      0.00014    132.433      0.000
## factor(race)1     -2.16846      0.10048    -21.582      0.000
## factor(race)2      2.21870      0.07695     28.833      0.000
## factor(race)7     -0.08810      0.13904     -0.634      0.527
## =====
## number of observations: 16351
## number of estimated parameters: 71
## residual degrees of freedom: 184
## maximum log-likelihood: -28977.06
##
## AIC(4): 58096.12
## BIC(4): 58642.97
## X^2(4): 733.5164 (Chi-square goodness of fit)
##
## ALERT: estimation algorithm automatically restarted with new initial values
##
```

```
lca_cov_w3$P
```

```
## [1] 0.1845714 0.1570558 0.6291794 0.0291934
```

```
lca_cov_w3$coeff
```

```
##              [,1]      [,2]      [,3]
## (Intercept)  1.6774836164  1.312215e+00 -0.018104014
## age          0.0004607971 -8.146898e-05  0.001647661
## factor(gender)2 -0.2034103090 -6.364798e-01 -0.689655047
## factor(degree)1 -1.9132967025  2.183320e-01 -1.898178111
## factor(degree)2 -1.3542716700  9.167995e-01 -1.449393586
## factor(degree)3 -3.9647104477  1.136569e+00 -2.208097621
## factor(degree)4 -3.5733138530  1.260583e+00 -2.594991206
## factor(degree)5 -1.9829481129  1.426268e+00 -1.355543607
## factor(degree)6 -3.1763371989  1.711636e+00 -1.915954043
## factor(degree)9 -0.0082289774 -2.523751e-02  0.018273829
## factor(race)1  -2.0866040840 -4.645438e-01 -2.168456965
## factor(race)2   2.9405243081  1.733186e+00  2.218699618
## factor(race)7   0.8365244604  6.234150e-02 -0.088096882
```